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A paleomagnetic pole position from the folded Upper Devonian Catskill red beds, and its tectonic implications

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ABSTRACT

Stable paleomagnetic components in the Upper Devonian Catskill red beds yield a pole position at lat 43.5°N, long 124.2°E (A_{95} = 4.5°). On the basis of results from a regional fold test on a collection of 54 samples from nine sites in the central Appalachians, we conclude that the magnetization predates folding and is therefore pre-Permian in age. This Catskill pole is similar to previously published Devonian poles obtained from flat-lying formations of cratonic North America. The positive fold test of the Catskill red beds is the first for rocks of Devonian age from the North American craton and eliminates the possibility that a Permian remagnetization occurred. Some samples, moreover, contained antiparallel characteristic magnetizations of both normal and reversed polarity, which further strengthens the arguments for a Late Devonian age of the magnetization. The Devonian poles of cratonic North America are, as previously noted by Kent and Opdyke, significantly different from Late Devonian poles of the New England-Canadian Maritime province area. This suggests the possibility that after Late Devonian time, the New England-Maritime province underwent sinistral strike-slip movements relative to cratonic North America.

INTRODUCTION

The Upper Devonian Catskill red beds and the time-equivalent Hampshire red beds extend from New York State to Virginia and are chiefly exposed in the Valley and Ridge province of the central Appalachian mountains (Colton, 1970; Cook and Bally, 1975). The 1,280-m-thick section in Pennsylvania consists dominantly of fluvial red micaceous sandstone and shale units, which are not correlatable laterally but are nevertheless widespread. Fine-grained hematitic sandstone in the section is suitable for paleomagnetic investigation. Kent and Opdyke (1978) published such a study on flat-lying red beds from southeastern New York State and noted that their pole at lat 46.8°N, long 116.7°E (δp = 6.1°, δm = 3.2°) agrees well with poles from Devonian limestone in Ohio (Martin, 1975) and Arizona (Elston and Bressler, 1977). These poles show, on the other hand, a significant discrepancy with poles from the New England area, which they interpreted in terms of relative movements. Because few Devonian poles were available for cratonic North America and because all were obtained from flat-lying formations, it became important to determine the age of the magnetizations. The purpose of this study, therefore, was to investigate the red sandstone from the folded Catskill red beds in the central Appalachians; a positive fold test would constrain the time of magnetization acquisition. Fifty-four samples were collected from nine sites in Pennsylvania, Maryland, and West Virginia.

RESULTS

Because many aspects of stratigraphy and similar magnetic properties have been amply discussed by Kent and Opdyke (1978), we will discuss only a few of the more salient details of our study.

Laboratory treatment of the samples involved standard procedures of stepwise demagnetization and subsequent analysis with Zijderveld (1967) diagrams and vector subtraction. Thermal, chemical, and alternating-field (AF) demagnetizations were performed, and all methods resulted in characteristic directions that were in excellent agreement. Demagnetization with AF techniques up to 100 mT alone generally isolated the characteristic direction

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but did not succeed in breaking down this component. Thermal demagnetizations isolated a characteristic direction by 450 °C, which subsequently unblocked by at most 670 °C. The grouping of directions greatly improved during demagnetization and generally resulted in site-mean directions with α_{95} values (Fisher, 1953) of 12° or less.

Representative Zijderveld diagrams of thermal and chemical demagnetizations are shown in Figure 1. Of special interest are the following phenomena: (1) the presence of a large component with low blocking temperature, moderate to low solubilities in HCl, and low coercivities; (2) the presence of antiparallel characteristic directions in some samples (for example, FPB 38); (3) the success of additional AF demagnetization with 10 mT during chemical leaching procedures (first done after 31 hr of cumulative leaching time); and (4) the variability in leaching time required to eliminate a substantial part of the characteristic magnetization. Although not very well defined, the in situ directions of the first-removed magnetization component generally conform to the present-day geomagnetic field direction and are thought to be of recent origin and, in part, of a viscous nature.

Figure 2 shows the characteristic sample directions and the site-mean directions, as listed also in Table 1. The site-mean directions show a strong improvement in grouping, upon correction for the tilt of the strata, with a corresponding increase of the precision parameter (k) from 12.3 to 108.6. The ratio of k_2/k_1, the precision parameters after and before tectonic correction, respectively, is 8.83 and indicates that the fold test is positive and significant at the 99% confidence level (McElhinny, 1964). Four samples showed both normal (N) and reversed (R) polarity characteristic directions, and both have been plotted in Figure 2. The R directions in these samples had lower blocking temperatures (for example, FPB 38 in Fig. 1) and were eliminated at 550 to 600 °C. In most of the samples, however, N directions were not detected even at temperatures as high as 665 °C (for example, FPB 51 in Fig. 1), and we conclude that the presence of N directions is not a general phenomenon in all samples. Yet other samples from sites 1, 2, and 9 suggested the presence of N directions superimposed on R directions, but without reaching a stable endpoint. In these samples, apparently the blocking-temperature spectra of the N and R components overlap to a large extent, if not completely. We note, however, that the presence of these components is quite obvious during thermal demagnetization (Fig. 3) and that by 560 °C the directions begin to fall in the northern hemisphere. Becuase of the simultaneous removal of the N and R components in most samples of site 1, only one characteristic direction (declination = 342°, inclination = −4°) could be obtained, and therefore no mean direction for site 1 is given in Table 1.

As was to be expected for these red beds, the characteristic directions of magnetization are found largely in hematite (aggregates, coating, and cement), as evidenced by thermal and chemical demagnetizations and by optical examinations. In view of the antiparallel (N and R) directions in some samples, we conclude that these directions could well represent a chemical remanent magnetization (CRM).

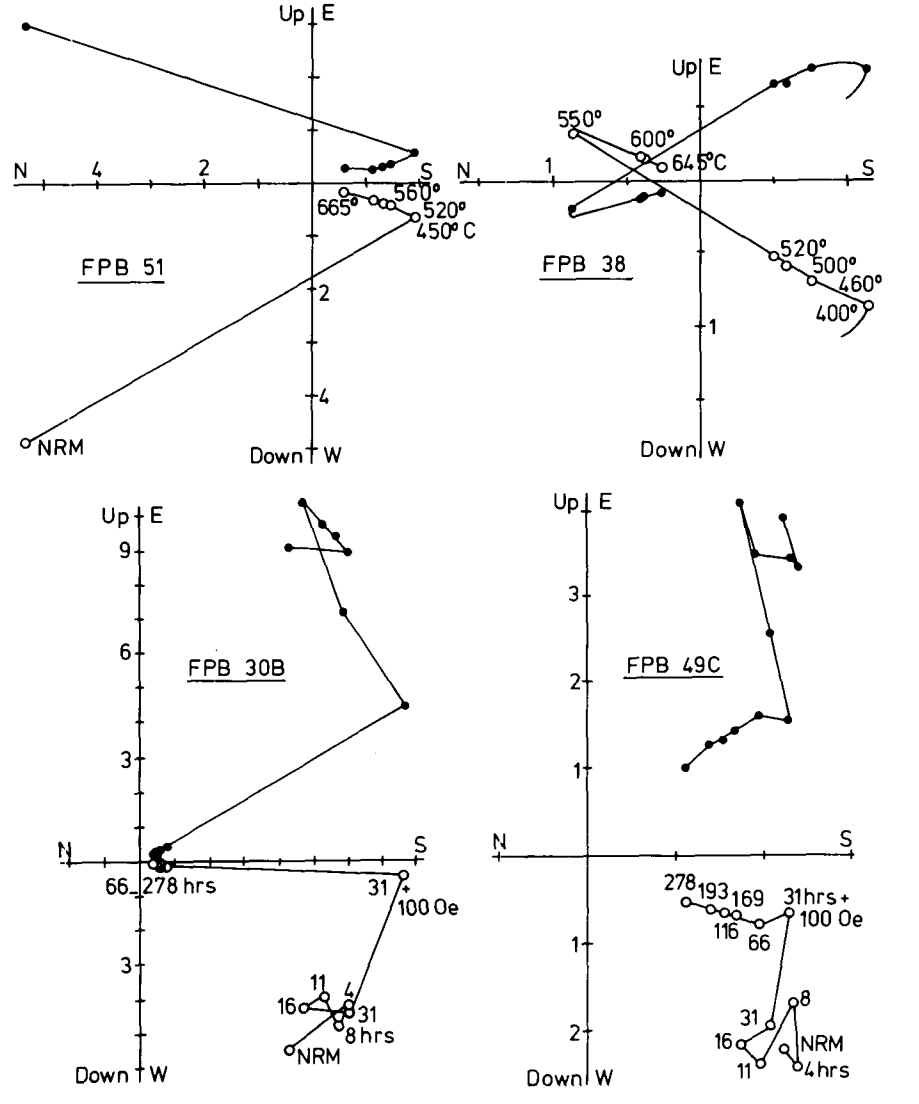


Figure 1. Orthogonal demagnetization (Zijderveld, 1967) diagrams of successive endpoints of magnetization vector during progressive thermal or chemical demagnetization, after correction for tilt of strata. Demagnetization temperatures (in degrees Centigrade) and cumulative leaching times (in hours) are indicated. Solid circles represent projections on horizontal plane, and open circles represent those on north-south vertical plane. Units indicated for axes are in milliamperes per metre (10^{-6} emu/cm^3). Samples from sites 7 (FPB 51), 5 (FPB 38), 4 (FPB 30B), and 6 (FPB 49C). Note that after 31 hr of leaching time (bottom diagrams) an additional 10 mT (100 Oe) alternating-field demagnetization has been applied.

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It is therefore of crucial importance to establish within reasonable limits that this CRM was acquired not long after deposition before we can make estimates of the paleoposition of North America during Late Devonian time. Two lines of evidence are in support of an age of the magnetization which is not much later than Late Devonian and certainly pre-late Carboniferous. The first is the success of the fold test, which points to a prefolding age for the magnetization. The age of the folding, although imprecisely known, is taken to be pre-Permian (Van der Voo, 1979). The second line of evidence is the occurrence of both normal and reversed polarities (Figs. 2, 3). The N directions, which were obtained from samples with stable endpoints, are perfectly antipodal to the R directions. This implies that the magnetizations of both polarities were acquired in a time short enough so that not much apparent polar wandering occurred. Because the field during the late-middle and late Carboniferous (as well as most of the Permian), being part of the Kiaman interval (Irving, 1964), was reversed, we argue that the magnetizations are pre–late Carboniferous in age.

DISCUSSION AND TECTONIC IMPLICATIONS

The mean pole position of this Catskill study is consistent with the Catskill pole of Kent and Opdyke (1978) at lat 46.8°N,

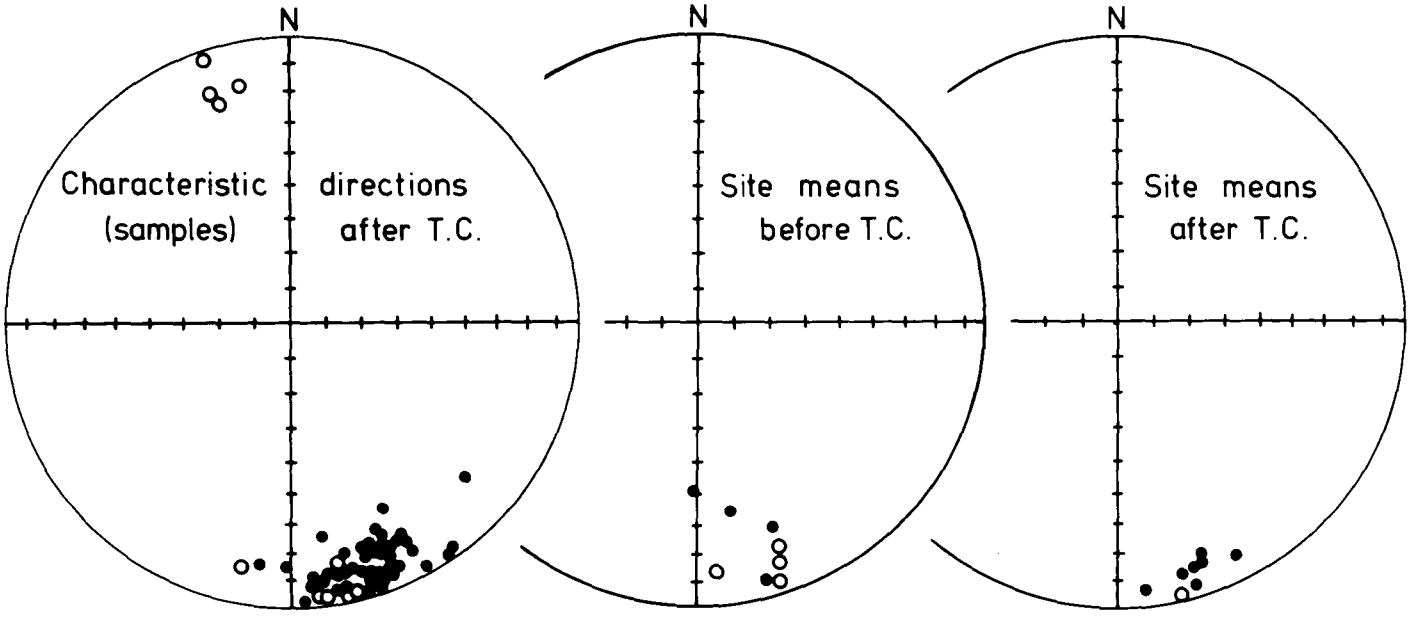


Figure 2. Equal-area projection of characteristic directions of magnetization (per sample) and of site-mean directions. Open symbols = projections on upper hemisphere; solid symbols = projections on lower hemisphere; T.C. = correction for tilt of strata.

TABLE 1. PALEOMAGNETIC DIRECTIONS AND POLE POSITIONS

Site	R_S*	Site coordinates °N	°W	Strike and dip (°)	Direction before T.C.† decl.	incl.	Direction after T.C.† decl.	incl.	k§	α_{95}§	Pole position °N	°E
1	0/4	39.29	78.57	40/70	No stable endpoint directions							
2	7/7	39.33	78.70	217/27	175.5	-11.9	174.3	+6.5	26.4	12.0°	47.1	109.7
3	3/3	39.29	78.60	43/54	181.6	+40.7	167.7	-1.7	229.4	8.2°	49.9	120.7
4	6/6	39.49	78.44	10/25	160.1	+23.8	153.6	+10.2	35.0	11.5°	39.2	136.4
5	8/9	39.71	78.36	205/33	161.4	-11.2	161.4	+11.7	24.5	11.4°	41.3	126.6
6	6/6	39.68	78.41	43/32	170.3	+32.3	163.9	+ 5.8	51.9	9.4°	44.9	124.7
7	6/6	39.97	78.24	219/36	160.6	-16.2	160.2	+14.8	119.9	6.1°	39.1	127.4
8	6/6	39.97	78.24	216/19	162.6	-4.3	163.2	+11.0	46.0	10.0°	41.9	124.5
9	7/7	40.44	77.95	184/29	165.1	+6.2	166.0	+9.8	134.3	5.2°	42.8	121.2
Mean 8/9					166.7	+7.3	163.8	+8.5			43.5	124.2
					k_1# = 12.3		k_2# = 108.6				k = 149.0	
					α_{95} = 16.4°		α_{95} = 5.3°				A_{95}** = 4.5°	

*Ratio of samples used to compute the mean to number of samples analyzed.
†Correction for the tilt of the strata; decl. and incl. are declination and inclination (in degrees).
§k and α_{95} are the precision parameter and the semiangle of the cone of 95% confidence, respectively (Fisher, 1953).
#k_1 and k_2 are the precision parameters before and after correction for the tilt of the strata, respectively.
**A_{95} is the cone of confidence for the mean pole, as calculated from the eight site-mean poles.

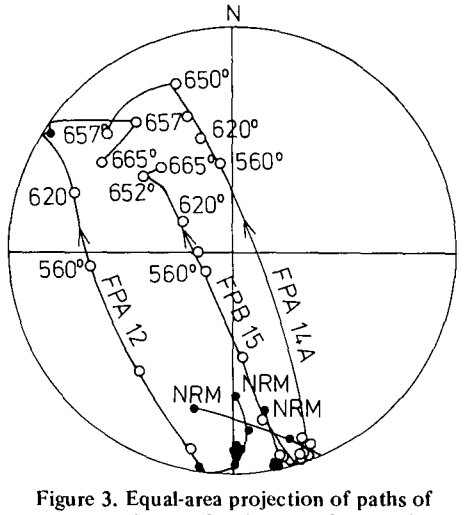


Figure 3. Equal-area projection of paths of successive positions of endpoints of magnetization vector during thermal demagnetization. Open circles = projections on upper hemisphere; solid circles = projections on lower hemisphere. Samples from site 2 (FPB 15) and site 9 (FPA 12 and 14A).

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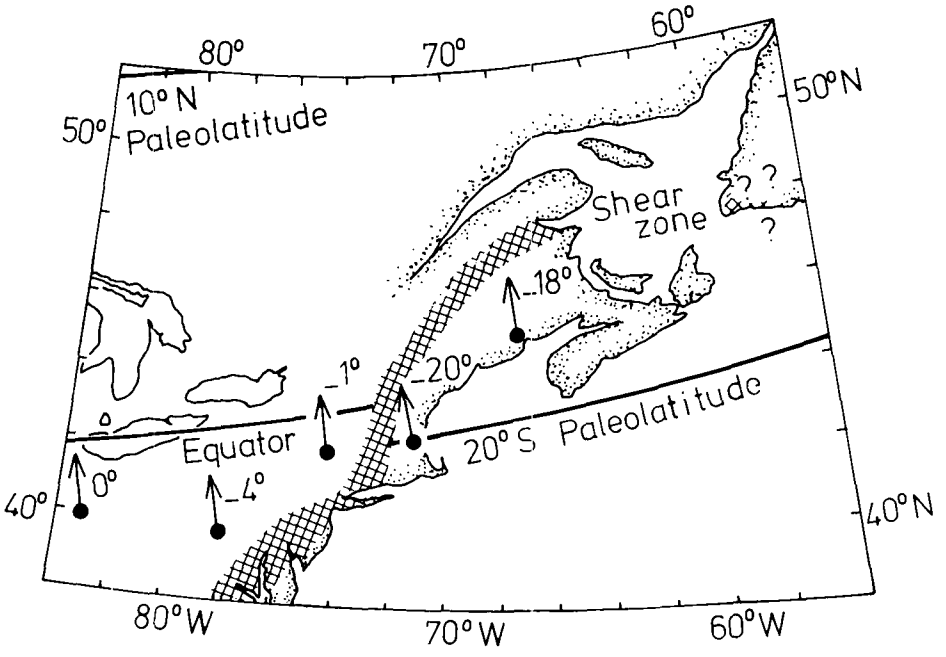


Figure 4. Map of eastern North America with paleodeclinations (arrows), paleolatitudes as derived from observed inclinations, and averaged paleolatitude lines for Late Devonian time. Cross-hatched shear zone is geologically most plausible demarcation zone between North American craton (left) and New England-Canadian Maritime province "prong" (right), after Kent and Opdyke (1978). Paleomagnetic results are, from left to right, from Martin (1975) for Ohio, from this study for southern Pennsylvania, from Kent and Opdyke (1978) for New York State, from Schutts and others (1976) for Massachusetts, and from Black (1964), Phillips and Heroy (1966), and Robertson and others (1968), combined, for Maine-New Brunswick.

long 116.7°E. Other poles from Ohio and Arizona are slightly older (Middle to early-Late Devonian), but they also plot nearby. On the other hand, four Late Devonian poles from the Perry Formation in Maine and New Brunswick (Black, 1964; Phillips and Heroy, 1966; Robertson and others, 1968) and one Devonian pole from low-grade metavolcanic rocks in Massachusetts (Schutts and others, 1976) are located about 20° to the south. We note that these paleomagnetic poles from the New England area do not all pass modern reliability criteria (such as demagnetization experiments, sufficient numbers of samples and sites, and full publication of experimental results). The pole by Robertson and others (1968) is probably one of the more reliable results in this group, and even for it one might argue that more laboratory studies are needed. Be that as it may, Kent and Opdyke (1978) have discussed the discrepancies between the Devonian and early Carboniferous poles from cratonic North America and the New England–Maritime province area, and have offered an elegant explanation in terms of relative motions between the two. We do not want to reiterate their arguments but instead have constructed a paleolatitude map of eastern North America to visualize their hypothesis (Fig. 4). It can be seen that the paleolatitude patterns derived from the paleomagnetic results of these two areas differ by about 18°, indicating a possible post–Late Devonian relative northward movement of the New England–Maritime province area. It should be noted that Late Devonian poles from Europe (Russian platform, Norway) are in excellent agreement with those from New England and New Brunswick and that the relative movements mentioned above are of megatectonic scope. They involve the North American craton, on the one hand, and cratonic Europe together with the recently formed Caledonide mountains and a New England–Maritime province "prong" on the other hand. The role of Newfoundland is unclear in this respect, as relevant Devonian paleomagnetic data are not available for this area. As suggested by Kent and Opdyke (1978), the relative sinistral movements occurred during the Carboniferous; they may well have been the result of the collision between Africa and the northern continents which produced the Hercynian-Alleghenian orogeny.

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